

Huvo AI — Forward Deployed Engineer Assignment

Deadline 24 hours

MANDATORY BACKEND: FastAPI (Python). Express.js or other backend frameworks will not be accepted.

Objective

Build a simple AI conversational bot for a fictional real-estate company.

The main focus of this assignment is prompt engineering and agent behaviour.

The prompt should be designed so that it can work across both chat and voice/calling interactions.

Scenario

You are building an AI sales agent for Northstar Homes.

Project	Northstar One
Location	Sector 79, Gurugram
Configurations	2 BHK and 3 BHK
Starting Price	2 BHK: ₹1.35 crore onwards 3 BHK: ₹1.75 crore onwards

The agent should communicate naturally in English, Hindi, and Hinglish.

Its goal is to understand the customer's requirements, answer relevant questions, qualify the lead, and help arrange a site visit.

Part 1 — Prompt

Create a strong system prompt for the AI agent.

The same prompt should be suitable for use in both chat and voice interactions.

It should handle:

- Natural conversation
- Customer qualification
- English, Hindi, and Hinglish
- Common objections
- Busy or uninterested customers
- Requests to contact later
- Requests to stop further communication
- Unknown questions
- Site-visit booking
- Booking failures
- Human escalation

- Proper conversation ending

The agent should not invent prices, discounts, availability, or other information that has not been provided.

Part 2 — Simple Bot

Build a simple text-based conversational bot with a simple web interface using your final prompt.

MANDATORY BACKEND: FastAPI (Python). Express.js or other backend frameworks will not be accepted.

The bot should:

- Accept customer messages
- Respond using your final prompt
- Remember information shared during the conversation
- Support English, Hindi, and Hinglish
- Handle different customer intents and objections
- Simulate a site-visit booking
- Handle a failed booking correctly
- After the conversation ends, generate analytics data from the conversation. Define useful fields such as budget, interest level, configuration, site-visit status, follow-up requirement, etc.
- Include a few test cases demonstrating that your bot behaves correctly across different scenarios. Show the input, expected behaviour, and actual output.

Keep the implementation simple. We are mainly interested in prompt quality, agent behaviour, and how you structure the solution.

Submission

1. Public GitHub Repository

Please share a public GitHub repository containing:

- Final prompt
- Source code
- README
- .env.example

Do not commit API keys, credentials, or other secrets.

2. Demo Video

Share a short Loom, Google Drive, or unlisted YouTube video showing:

- The bot working
- A sample conversation
- Your prompt approach
- A brief explanation of the implementation

3. README

Include:

- How to run the bot
- Key assumptions
- Known limitations
- AI tools used, if any

4. Send Your Submission

Once completed, please email your assignment to:

To aditi@huvo.ai
CC nikhil@huvo.ai, vaibhav@huvo.ai, rohit@huvo.ai

Please include:

- Public GitHub repository link
- Demo video link
- Any relevant notes about the submission

Evaluation

We will mainly evaluate:

- Prompt quality
- Agent behaviour
- Handling of different customer situations
- Conversation context and memory
- Whether the bot works
- Code clarity
- Your understanding of the solution

Selected candidates will be invited for an interview.